MVP

MVP Document: ShopForMe

Project Title: ShopForMe

Date: May 21, 2024

Version: 1.1

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1. Project Overview

Objective: Enhance the shopping experience for the blind or visually impaired by providing a mobile application that communicates with IoT devices to assist users in navigating stores.

Problem Statement: Navigating a store can be challenging for blind or visually impaired individuals due to the difficulty in locating specific items.

Proposed Solution: A mobile application that uses voice control to manage shopping lists and provides audio feedback with precise directions to item locations using IoT technology.

Expected Benefits: Greater independence for visually impaired shoppers and a more accessible shopping environment.

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2. MVP Objectives

- **Primary Objectives:**
  - Implement voice-controlled list management and product search.
  - Provide accurate real-time navigation inside stores.
  - Deliver audio feedback for directions and aisle confirmation.
- **Success Criteria:**
  - Precision in location and direction information.
  - User satisfaction with the voice interaction model.
  - Effective audio feedback that enhances user navigation.

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3. Personas and Use Cases

Target Users:

- **Persona 1:** John Doe, visually impaired, frequent smartphone user.
- **Persona 2:** Jane Smith, blind, active technology user for accessibility.

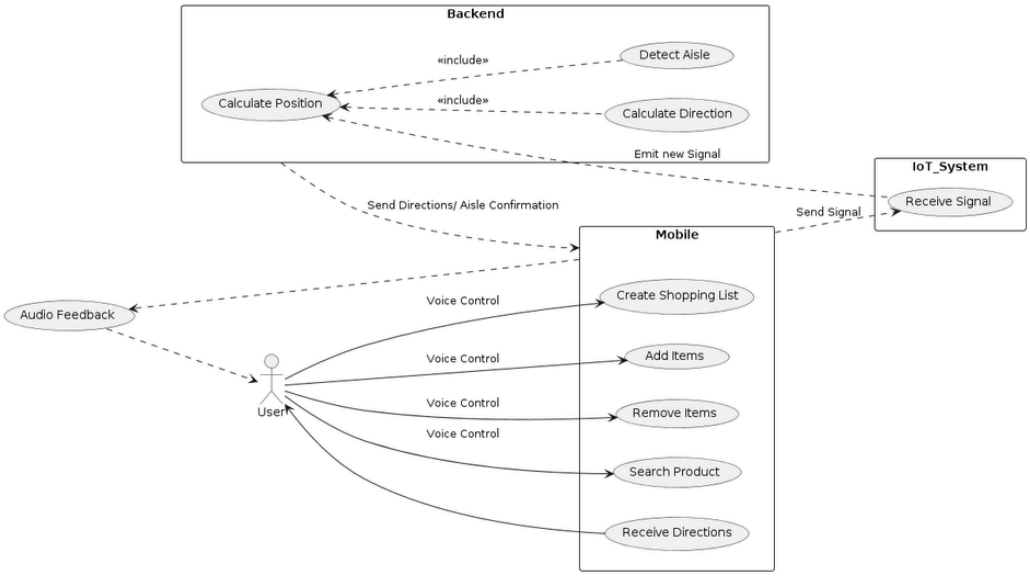
Use Case:

- **Scenario:** Jane Smith wants to find pasta in the supermarket. She uses the app to add "pasta" to her shopping list with voice commands. The app, interacting with in-store IoT devices, guides her directly to the pasta aisle with audio directions.

Use Case:

- Code schéma

```
@startuml
left to right direction
actor User

rectangle Mobile {
  User --> (Create Shopping List):Voice Control
  User --> (Add Items):Voice Control
  User --> (Remove Items):Voice Control
  User --> (Search Product):Voice Control
  User <-- (Receive Directions)
}

rectangle IoT_System {
  Mobile ..> (Receive Signal):Send Signal
}

rectangle Backend {
  (Calculate Position) <.. (Receive Signal): Emit new Signal
  (Calculate Position) <.. (Calculate Direction): <<include>>
  (Calculate Position) <.. (Detect Aisle): <<include>>
}

Backend ..> Mobile:Send Directions/ Aisle Confirmation
Mobile ..> (Audio Feedback)
(Audio Feedback) ..> User

@enduml


```



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4. Key Features

- **Voice Control**: Users can interact with the app using voice commands to manage their shopping list.
- **IoT Integration**: IoT devices in the store receive signals from the mobile app to help locate the user.

- **Real-Time Navigation**: Backend algorithms calculate the user's position and send navigational directions to the mobile app.
- **Audio Feedback**: Directions and aisle confirmations are provided audibly to assist users in finding items.

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5. Technical Specifications

- **Technologies Used:**
  - **IoT Devices:** Equipped with Bluetooth Low Energy.
  - **Mobile Application:** Built with Flutter for cross-platform compatibility.
  - **Backend:** ?
- **Data Flow:**
  - The user interacts with the mobile app through voice commands.
  - The mobile app sends signals to IoT devices.
  - IoT devices relay location data to the backend.
  - The backend processes this data, calculates directions, and sends them back to the mobile app.
  - The user receives these directions as audio feedback.

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6. System Architecture

This section refers to the provided UML diagram, detailing each component's role and interaction.

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7. Project Team

- **Project Manager:** @Frederic LE DOUARIN
- **Mobile Developer:** @Rémi Perray, @Corentin Ribeyron
- **IoT Developer: @Franques Matthieu, @Maxime Delmon, @Elhoucine Haddouch**
- **Backend Developer:** @Franques Matthieu, **@Maxime Delmon, @Elhoucine Haddouch**
- **QA Engineer:** ?

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8. Risks and Constraints

- **Potential Risks:**
  - Delays in signal transmission could affect real-time response.
  - Challenges in achieving high accuracy in positioning within certain store layouts.
- **Constraints:**
  - Dependency on the store's infrastructure for IoT device placement.
  - Budget constraints for extensive IoT setup.

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9. Glossary

- **IoT:** Internet of Things, referring to the network of physical devices that gather and communicate data.
- **BLE:** Bluetooth Low Energy, used for short-range communication between devices.
- **Flutter:** A framework for building native apps using React